Vidyalankar Institute of Technology
Semester VII – Information Technology - Mid Semester Assessment – 2

Date: 05/10/2026	Natural Language Processing	PC_PEC/ PE-6	20 Marks/45 mins.

<table>
<tr><th>1</th><th colspan="2">Solve any two (5 marks each)</th><th>CO</th></tr>
<tr><td></td><td>A</td><td>Explain Lexical semantics.
For each pair, identify the semantic relation:
i. car – vehicle
ii. branch – tree
iii. rise – fall
iv. fare – price</td><td>CO4</td></tr>
<tr><td></td><td>B</td><td>Why is Word sense disambiguation important?
Disambiguate the sense of word “bank” using Lesk algorithm:
Target word: bank
Context:
The bank accepted my deposits and approved my mortgage.
Two possible senses:
bank₁: financial institution, deposits, lending, mortgage
bank₂: sloping land beside a river, canoe, water, currents</td><td>CO3</td></tr>
<tr><td></td><td>C</td><td>A large e-commerce company wants to develop an NLP-based search system that understands relationships between product-related words, including frequently occurring and rare terms. As an NLP engineer, compare the CBOW and Skip-gram Word2Vec models and recommend a suitable model for this application. Explain your reasoning with suitable diagrams.</td><td>CO5</td></tr>
<tr><td>2</td><td colspan="2">Solve anyone (10 marks each)</td><td></td></tr>
<tr><td></td><td>A</td><td>Based on the given state transition and emission probability matrix, assign POS to the statement: That Girl Smiles
Emission Probability Matrix
<table><tr><th>Word</th><th>DT</th><th>NN</th><th>VB</th></tr><tr><td>That</td><td>0.40</td><td>0.00</td><td>0.00</td></tr><tr><td>Girl</td><td>0.00</td><td>0.015</td><td>0.0031</td></tr><tr><td>Smiles</td><td>0.00</td><td>0.0004</td><td>0.20</td></tr></table>State Transition Matrix
<table><tr><th>Previous Tag</th><th>DT</th><th>NN</th><th>VB</th></tr><tr><td><S></td><td>0.50</td><td>0.40</td><td>0.10</td></tr><tr><td>DT</td><td>0.01</td><td>0.99</td><td>0.00</td></tr><tr><td>NN</td><td>0.30</td><td>0.30</td><td>0.40</td></tr><tr><td>VB</td><td>0.40</td><td>0.40</td><td>0.20</td></tr></table></td><td>CO3</td></tr>
<tr><td></td><td>B</td><td>An NLP-based question-answering system encounters sentences containing pronouns whose references are ambiguous. Explain how Hobbs Algorithm can resolve such references. Apply the algorithm to following statement and represent the process using a parse tree.
John has a new Acura. He gave it to Mary.</td><td>CO3</td></tr>
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CO3	Apply parsing algorithms and semantic techniques (like WSD and discourse resolution) to analyze linguistic structures.
CO4	Analyze the structure and relationships in lexical resources (e.g., WordNet, BabelNet) and evaluate semantic similarity and ambiguity.
CO5	Evaluate the performance of various neural language models and word embedding techniques on language tasks.